NON-EXTRACTIVE CAPITAL

Digital Sovereignty Fund

Unified Financial, Impact, and Theological Model

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Consolidated working paper

Science Park 608 1098 XH
Amsterdam, The Netherlands

info@nonextractivecapital.com
+31 (0) 20 2621255

Contents

Science Park 608 1098 XH
Amsterdam, The Netherlands
info@nonextractivecapital.com
+31 (0) 20 2621255

1 Executive Overview

The model is built in layers.

1. The **financial layer** explains how a portfolio with asymmetric deployment and capped repayment can produce sustainable returns.
2. The **impact layer** explains how that capital becomes sovereign open-source digital infrastructure rather than merely financial exit value.
3. The **theological layer** asks what kind of return is morally legitimate, decomposes the repayment claim, and then feeds that moral structure back into both finance and impact.

The central intuition is simple. Conventional venture capital tends to maximize return by inflating upside extraction. The DSF logic instead aims to improve survival, durability, openness, sovereignty retention, and reinvestment, while morally constraining the claims of capital.

In compact form, the three core equations are:

$$M = \frac{rkp}{1 + (k-1)p}, \tag{1}$$

$$I = N \cdot p \cdot L \cdot o \cdot d \cdot a \cdot e, \tag{2}$$

$$T = 1 - U + \mu\eta. \tag{3}$$

Here:

- M is the single-cycle financial multiple,
- I is the infrastructure impact outcome,
- T is theological integrity,
- p is survival probability,
- r is the repayment cap,
- k is capital concentration into companies that survive,
- U is usury pressure,
- η is the reinvestment ratio.

The full design objective is not to maximize one number. It is to maximize impact while remaining both financially viable and theologically legitimate:

$$\max I \quad \text{subject to } F \geq F_{\min} \text{ and } T \geq T_{\min}.$$

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Amsterdam, The Netherlands
info@nonextractivecapital.com
+31 (0) 20 2621255

2 Financial Portfolio Model

This section rebuilds the financial model slowly from first principles. The aim is not only to present the final equation, but also to explain why the equation takes this shape, what strategic decisions each variable represents, and how the closed-form model later connects to an operational evergreen fund.

2.1 Why the financial model starts with survival rather than exits

A conventional venture model is usually narrated through rare outsized exits. The DSF logic starts elsewhere. It assumes that portfolio construction, governance discipline, and mission-aligned support should raise the proportion of companies that remain alive long enough to repay capped amounts. In other words, the model is built around *survival and disciplined repayment*, not around the hunt for unlimited upside.

That is why the core variables are:

- p: the probability that a portfolio company survives long enough to become a meaningful repayer,
- k: the extent to which capital is concentrated into companies that show evidence of survival,
- r: the capped repayment multiple that a successful company may pay back.

The final closed-form equation

$$M = \frac{rkp}{1 + (k - 1)p}$$

can therefore be read as a compact summary of a strategic thesis: *improve the quality and durability of the portfolio, concentrate capital into the companies that prove viable, and cap what capital may claim back from success.*

2.2 Step 1: Begin with company counts

The cleanest starting point is not probabilities but a finite portfolio.

Let:

$$\begin{aligned} N &= \text{total number of companies funded,} \\ S &= \text{number of successful companies,} \\ F &= \text{number of failed companies.} \end{aligned}$$

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Then:

$$F + S = N.$$

At this stage, nothing theological or impact-related has been added yet. We are simply asking: if a fund backs N companies, and some fail while others survive, how much capital goes out and how much comes back?

2.3 Step 2: Add asymmetric deployment

The next move is essential. DSF does not assume that every company receives the same amount of capital. That would be both financially unrealistic and strategically clumsy. Instead, the model assumes a smaller initial exposure to companies that may fail, followed by larger follow-on deployment into companies that demonstrate traction and survival.

Define:

$$I_f = \text{capital committed to a failure-stage company},$$
$$I_s = \text{capital committed to a successful company}.$$

To express the asymmetry compactly, introduce:

$$I_s = kI_f$$

where k is the success investment multiplier. If $k = 5$, then a company that proves itself may receive five times as much capital as a company that does not progress beyond the early stage.

This asymmetry matters because it allows the fund to place limited downside exposure on weak performers while still putting meaningful capital behind the companies that look durable. In plain English: the model is trying to lose small and win steadily.

2.4 Step 3: Compute total capital out and total capital back

Once the asymmetric deployment rule is in place, the total capital invested across the portfolio is:

$$\text{Investment} = I_f F + I_s S.$$

Substituting $I_s = kI_f$ gives:

$$\text{Investment} = I_f F + kI_f S = I_f(F + kS).$$

Now introduce the repayment cap. If a successful company repays at a capped multiple r, then total repayment is:

$$\text{Repayment} = rI_sS = rkI_fS.$$

This is the first point at which the structure of the fund becomes visible. The amount returned to the fund depends on three things at once:

- how many companies survive (S),
- how much extra capital was concentrated into them (k),
- and what repayment cap applies to that capital (r).

2.5 Step 4: Turn the portfolio into a multiple

A portfolio multiple is simply money back divided by money out:

$$M = \frac{\text{Repayment}}{\text{Investment}} \quad (4)$$

$$= \frac{rkI_fS}{I_f(F + kS)} \quad (5)$$

$$= \frac{rkS}{F + kS}. \quad (6)$$

This is already a useful form. It shows that the fund does not need every company to succeed. What matters is the relationship between the number of successes and the total capital committed across failures and successes together.

Substituting $F = N - S$ yields:

$$M = \frac{rkS}{N + (k-1)S}.$$

This expression makes the deployment logic more visible. When S rises, the numerator rises, but so does the denominator because more capital has also been concentrated into the companies that survived.

2.6 Step 5: Convert counts into a success probability

Because portfolio analysis is easier to compare across different fund sizes when stated in rates rather than raw counts, define:

$$p = \frac{S}{N}.$$

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Substituting $S = pN$ into the previous expression and simplifying gives the compact DSF equation:

$$M = \frac{rkp}{1 + (k-1)p}. \tag{7}$$

This equation is the financial core of the whole document. It says:

- **If survival improves** ($p \uparrow$), portfolio performance rises.
- **If more capital is reserved for proven companies** ($k \uparrow$), portfolio performance can rise, though only if enough companies survive.
- **If the repayment cap rises** ($r \uparrow$), performance rises mechanically - but this is exactly the variable later subjected to theological discipline.

The denominator,

$$1 + (k-1)p,$$

is the crucial balancing term. It prevents the model from pretending that greater concentration into winners is free. If more companies survive, the fund has also committed more follow-on capital into those companies.

2.7 Step 6: Read the strategic meaning of the variables

At this stage, it is worth pausing to interpret the equation strategically.

The variable p is the most mission-consistent path to stronger performance.

Improving p means building a portfolio that lasts: better governance, better procurement readiness, better ecosystem support, better founder-company fit, and better alignment between mission and capital structure.

The variable k reflects capital discipline.

A higher k means the fund keeps early losses smaller and reserves more serious capital for the firms that demonstrate viability. But an excessively high k can also make the portfolio brittle if survival is weak.

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The variable r is the most financially tempting but morally sensitive lever.

Raising r improves the multiple immediately, but later sections show that not every way of justifying a higher r is morally equivalent. This is where theology will enter the model.

2.8 Step 7: Benchmark intuition - what does a 5% annual VC-equivalent return imply?

In an evergreen structure, the easiest first comparison is the multiple corresponding to a steady annual return over different horizons:

$$A = P(1 + r)^t.$$

For a 5% annual return, the implied cumulative multiple is:

Years	Multiple
3	1.16×
5	1.28×
7	1.41×
10	1.63×
12	1.80×
15	2.08×
20	2.65×

This table matters because a capped repayment multiple that looks modest in isolation may still compare well to ordinary benchmark expectations once time is taken seriously.

A 3× repayment looks very different depending on how long it takes to arrive:

Repayment	Years	Annual Return
3×	10 years	≈ 11.6%
3×	15 years	≈ 7.6%
3×	20 years	≈ 5.6%

The point is not that every company should repay at 3× quickly. The point is that a capped, non-extractive repayment model can still produce financially respectable outcomes when the fund is patient and evergreen.

2.9 Step 8: See the logic in stylised portfolio examples

The original financial exploration began with stylized 25-company portfolios where failed companies receive less capital than successful ones. These examples are useful because they make the general equation tangible.

Case A: €25k into failures and €100k into successes.

Investment and repayment are:

$$\text{Investment} = 25{,}000F + 100{,}000S, \qquad \text{Repayment} = 300{,}000S$$

assuming a 3× repayment cap for successful companies.

This gives the following illustrative outcomes:

Successes	Failures	Capital Invested	Repayment	Multiple
2	23	€775k	€600k	0.77×
3	22	€850k	€900k	1.06×
10	15	€1.375M	€3.0M	2.18×
15	10	€1.875M	€4.5M	2.40×

Approximate 10-year annual returns for two of those multiples are:

Portfolio Multiple	Annual Return
2.18×	≈ 8.1%
2.40×	≈ 9.2%

The lesson from Case A is that a portfolio does not need a heroic number of successes to cross into attractive return territory. Once enough companies survive to justify the follow-on concentration, capped repayments can do meaningful work.

Case B: €50k into failures and €250k into successes.

Now the model becomes:

$$\text{Investment} = 50{,}000F + 250{,}000S, \qquad \text{Repayment} = 750{,}000S.$$

Illustrative scenarios:

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+31 (0) 20 2621255

Scenario	Capital Invested	Repayment	Multiple
20% success (5 successes)	€2.25M	€3.75M	1.67×
40% success (10 successes)	€3.25M	€7.5M	2.31×
60% success (15 successes)	€4.25M	€11.25M	2.65×

Approximate 10-year annual returns:

Scenario	Annual Return
20% success	≈ 5.3%
40% success	≈ 8.7%
60% success	≈ 10.2%

Break-even against a 10-year 5% benchmark is roughly 1.63×, which in this stylized 25-company case implies approximately 5 successes out of 25, or a 20% success rate.

These worked examples explain why the DSF thesis gives so much attention to survival and selective follow-on deployment. A moderate success rate can already be enough when losses are bounded and later capital is concentrated into the companies that prove durable.

2.10 Step 9: Present the generalized portfolio model

The financial model becomes much cleaner once the funding asymmetry itself is parameterised.

Define:

$$F + S = N, \tag{8}$$

$$I_f = \text{investment in a failure-stage company}, \tag{9}$$

$$k = \text{success investment multiplier}, \tag{10}$$

$$I_s = kI_f. \tag{11}$$

Then total capital invested is:

$$\text{Investment} = I_f F + I_s S = I_f(F + kS).$$

Let r be the repayment multiple on successful companies. Total repayment is:

$$\text{Repayment} = rI_s S = rkI_f S.$$

So the single-cycle portfolio multiple is:

$$M = \frac{\text{Repayment}}{\text{Investment}} \tag{12}$$
$$= \frac{rkI_f S}{I_f(F + kS)} \tag{13}$$
$$= \frac{rkS}{F + kS}. \tag{14}$$

Substituting $F = N - S$ gives:

$$M = \frac{rkS}{N + (k-1)S}.$$

Introducing the success rate

$$p = \frac{S}{N}$$

produces the most compact form of the model:

$$M = \frac{rkp}{1 + (k-1)p}. \tag{15}$$

This equation contains the core portfolio logic:

- higher p improves returns,
- higher k concentrates more capital into companies that survive,
- higher r raises the repayment claim,
- but the denominator captures the cost of concentrated follow-on deployment.

2.11 Step 10: Move from one cycle to an evergreen fund

A one-cycle multiple is not yet an evergreen fund. The next step is to ask what happens when repaid capital is not simply distributed and terminated, but returned to the fund and redeployed.

If capital is recycled across c investment cycles, the cumulative multiple becomes:

$$M_{\text{total}} = M^c = \left(\frac{rkp}{1 + (k-1)p}\right)^c. \tag{16}$$

This is the simplest evergreen extension of the portfolio model. It shows how even moderate single-cycle multiples can produce substantial long-run outcomes if capital is repeatedly redeployed.

Conceptually, this step changes the story of the fund. Instead of:

$$\text{capital} \longrightarrow \text{companies} \longrightarrow \text{exit},$$

the evergreen structure says:

$$\text{capital} \longrightarrow \text{companies} \longrightarrow \text{repayment} \longrightarrow \text{new companies}.$$

That change is already financially important. Later it will also become theologically important.

2.12 Step 11: Compare evergreen performance against a benchmark

If one compares the evergreen model against a benchmark equivalent to a 5% annual VC return, and assumes 10 years per cycle, then:

$$Y = 10c, \qquad M_{\text{VC}} = (1.05)^{Y} = (1.05)^{10c}.$$

The model passes the benchmark whenever:

$$M_{\text{total}} > M_{\text{VC}}.$$

For $k = 5$ and $r = 3$, example outcomes are:

p	c	M_{total}	PASS / FAIL
0.10	1	1.15	FAIL
0.20	1	1.67	PASS
0.40	1	2.31	PASS
0.60	1	2.65	PASS
0.10	2	1.33	FAIL
0.20	2	2.78	PASS
0.40	2	5.34	PASS
0.60	2	7.02	PASS
0.10	3	1.53	FAIL
0.20	3	4.64	PASS
0.40	3	12.34	PASS
0.60	3	18.62	PASS

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For other levels of capital concentration at $r = 3$:

Scenario	$p = 0.20$	$p = 0.40$	$p = 0.60$
$k = 3$	1.50×	2.00×	2.25×
$k = 8$	1.91×	2.67×	3.09×

These tables help the reader see the fund's real strategic choice. A conventional investor may try to push performance by stretching r. DSF can also push performance by improving p or by choosing a more disciplined k.

2.13 Step 12: Translate the algebra into an operational cash-flow model

The closed-form portfolio model is useful, but an operating fund must also pay people, cover administration, potentially service debt, and decide how much to distribute versus recycle. That is why the spreadsheet-style cash-flow extension matters.

Average annual inflow over a repayment horizon T can be written as:

$$R_{\text{year}} = \frac{rkI_f S}{T}.$$

Operating cost can be represented as:

$$C_{\text{ops}} = \text{salary} + \text{admin}.$$

Management fees can be written as:

$$C_{\text{mgmt}} = m \times \text{Fund}.$$

A stylised annual loan repayment can be written as:

$$\text{Loan}_{\text{year}} = \frac{L(1+i)}{T_L}$$

where L is principal, i is the financing rate, and T_L the amortisation horizon.

Net annual cash flow is then:

$$\text{Cash}_{\text{year}} = R_{\text{year}} - C_{\text{ops}} - C_{\text{mgmt}} - \text{Loan}_{\text{year}}.$$

If a fraction d_{LP} is distributed to LPs, then the recyclable remainder is:

$$\text{Recycle} = (1 - d_{\mathrm{LP}})\,\text{Cash}_{\text{year}}.$$

This is the point at which the document naturally transitions from closed-form algebra to spreadsheet simulation. The closed-form model explains *why the fund can work in principle*. The annual cash-flow model explains *how the fund remains liquid and operational in practice*.

2.14 Step 13: What the financial model is really optimizing

At the end of the financial build-up, the reader should be able to see that the model is not secretly trying to recreate conventional VC with softer language. Its inner logic is different:

- it prefers improving survival over inflating extraction,
- it uses capital concentration as discipline rather than as speculation,
- it treats capped repayment as a sustainable engine rather than a consolation prize,
- and it works especially well when combined with evergreen recycling.

This is the financial foundation on top of which the impact model is built.

3 Impact Model and Theory of Change

The financial model explains how capital can be returned. The impact model explains what that capital is meant to create in the world. This section rebuilds the impact equation step by step so that the reader can see why the outcome variable is not valuation, but durable sovereign digital infrastructure.

3.1 Why impact needs its own equation

A fund can be financially successful and still fail at its mission. For DSF, impact cannot be reduced to company valuation, exit size, or even gross revenue. The purpose of the vehicle is to create and preserve strategically important digital infrastructure under conditions of openness, stewardship, and European control.

That is why the document introduces a distinct impact equation:

$$I = N \cdot p \cdot L \cdot o \cdot d \cdot a \cdot e.$$

The logic is cumulative. Impact grows when:

- more relevant projects are funded (N),
- more of them survive (p),
- they remain alive for longer (L),
- they stay open (o),
- they remain sovereign and European (d),
- they are actually adopted (a),
- and they catalyse further ecosystem effects (e).

3.2 Step 1: Begin with one surviving company

Imagine first a single company that survives and remains operational. Even then, the fund's mission is not yet fulfilled. A company matters only to the extent that it remains useful, open, sovereign, adopted, and generative.

A simple one-company impact expression is therefore:

$$I_{\text{one}} = L \cdot o \cdot d \cdot a \cdot e.$$

This expression already contains the DSF worldview:

- a company that dies quickly has low impact even if it once looked promising,
- a company that closes its code reduces shared infrastructure value,
- a company that migrates governance or ownership away from Europe weakens sovereignty,
- a company that nobody adopts does not become infrastructure,
- and a company that creates no spillovers remains an isolated firm rather than an ecosystem node.

3.3 Step 2: Move from one company to expected portfolio successes

The fund does not back one company. It backs a portfolio.

If N projects are funded and the survival probability is p, then the expected number of durable surviving companies is:

$$N \cdot p.$$

At this point, the impact model and the financial model intersect directly. The same variable p matters in both places. Higher survival means more repayment potential *and* more infrastructure creation.

3.4 Step 3: Add lifetime

The first multiplication after survival is lifetime:

$$I \propto N \cdot p \cdot L.$$

This term matters because infrastructure is not a momentary event. It is a durable condition. A company that survives for ten years while maintaining useful technology contributes much more to sovereignty than a company that burns brightly and disappears.

In practical terms, L captures the difference between:

- short-lived startup activity,
- and durable institutional presence.

3.5 Step 4: Add openness retention

Digital sovereignty in this framework is not merely about geography. It is also about whether technology remains genuinely open.

Introduce:

$$o = \text{openness retention factor.}$$

If a company remains fully open-source over time, o stays close to 1. If it drifts toward enclosure, proprietary gating, or an "open-core" extraction pattern, o declines.

This is a crucial modelling choice. The fund is not counting any software company as equal impact. It is explicitly distinguishing shared digital infrastructure from technology that becomes privately fenced.

3.6 Step 5: Add sovereignty retention

A technology can stay open and still cease to serve European digital sovereignty if control, governance, or strategic IP migrate elsewhere.

Introduce:

$$d = \text{sovereignty retention factor.}$$

If governance, stewardship, and IP remain embedded in the intended community, then d remains high. If the company's decision rights drift toward external or extractive control, then d falls.

This is where the model becomes more than a generic impact framework. It is not simply measuring whether software exists. It is measuring whether strategically meaningful digital capacity remains located in the community the fund is meant to serve.

3.7 Step 6: Add adoption

A project can survive, remain open, and stay European, yet still have very little public significance if nobody uses it. Infrastructure must be *taken up*.

Introduce:

$$a = \text{adoption factor.}$$

This term captures deployment, usage, integration, and practical reliance. It moves the model from producer-side existence to user-side significance.

This is especially important in digital sovereignty work. A technically admirable open-source project does not become sovereign infrastructure merely because it was built. It becomes infrastructure when institutions, companies, developers, or public bodies actually depend on it.

3.8 Step 7: Add ecosystem spillovers

Finally, the model adds:

$$e = \text{ecosystem spillover multiplier.}$$

This term captures the fact that some projects do more than survive and get used. They also:

- attract contributors,
- enable forks and extensions,
- create standards and shared tooling,
- support adjacent ventures,
- and strengthen an entire ecosystem rather than a single company alone.

Science Park 608 1098 XH
Amsterdam, The Netherlands
info@nonextractivecapital.com
+31 (0) 20 2621255

Once this term is added, the full impact equation appears:

$$I = N \cdot p \cdot L \cdot o \cdot d \cdot a \cdot e. \tag{17}$$

3.9 Step 8: Why the variables multiply rather than add

The multiplicative form is important. It says the dimensions of impact are not independent substitutes for one another. Extremely strong performance on one dimension does not fully compensate for collapse on another.

For example:

- high adoption with low sovereignty retention can still undermine the mission,
- high survival with low openness can still produce enclosure rather than shared infrastructure,
- and high openness with no adoption creates little strategic effect.

A multiplicative structure therefore fits the mission better than a simple additive scorecard. It encodes the idea that sovereign digital infrastructure requires several conditions to hold together at once.

3.10 Step 8A: Why the variables are equally weighted for now

At this stage of the model, the variables in

$$I = N \cdot p \cdot L \cdot o \cdot d \cdot a \cdot e$$

are all given equal weight. This is a simplifying assumption used for clarity and illustrative purposes.

The equal-weighted form is useful because it keeps the logic transparent. In the absence of portfolio data, it allows the reader to see the full set of conditions that must jointly hold for sovereign digital infrastructure to emerge, without prematurely implying a precision that the model cannot yet justify.

At the same time, the equal-weighted form should not be interpreted as a final empirical claim that each variable has exactly the same proportional influence on impact. In practice, some variables may matter more than others, some may behave more like threshold conditions than smooth contributors, and some may partially overlap causally.

In particular:

- N and L are scale variables,
- p, o, d, and a are bounded factors or retention rates,
- and e is a spillover multiplier whose magnitude may be harder to estimate directly.

This means that a more realistic future specification may eventually use:

- normalisation terms,
- unequal exponents or weights,
- and possibly threshold or gating functions for variables such as openness and sovereignty retention.

One possible future form is therefore:

$$I = I_0 \cdot \left(\frac{N}{N_{\text{ref}}}\right)^{\alpha} \cdot p^{\beta} \cdot \left(\frac{L}{L_{\text{ref}}}\right)^{\gamma} \cdot a^{\delta} \cdot e^{\varepsilon} \cdot g(o, d), \tag{18}$$

where

$$g(o, d) = \begin{cases} 0, & o < \underline{o} \text{ or } d < \underline{d}, \\ o^{\zeta} d^{\eta}, & \text{otherwise.} \end{cases}$$

This more nuanced formulation is not yet used in the present document, precisely because the fund does not yet have realised portfolio performance data from which such weights or thresholds could be estimated or calibrated responsibly.

For that reason, the equal-weighted multiplicative model is retained here as the current working form. It is simple, transparent, and appropriate for an ex ante theory-of-change model. As DSF accumulates evidence from the performance of its actual portfolio, the impact equation can later be revised into a more differentiated and empirically informed specification.

3.11 Step 9: Recognize the structural guarantees

The DSF impact framework is deliberately structural rather than KPI-first. Portfolio companies must satisfy three entry conditions:

- steward ownership,
- open-source technology,
- European retention of governance and IP.

These are not merely downstream metrics. They are the architecture of the portfolio itself.

Because steward ownership, open-source licensing, and European retention are portfolio requirements, the framework often allows a simplified interpretation:

$$o \approx 1, \qquad d \approx 1, \qquad I \approx N \cdot p \cdot L \cdot a \cdot e.$$

This is one of the deepest ideas in the whole model: impact is not something the fund hopes for after the fact. It is hardcoded into portfolio construction.

3.12 Step 10: Explain the theory of change in prose

The underlying theory of change is:

1. Europe produces digital innovation, but the dominant venture financing model is organised around rapid exits.
2. That financing model places pressure on companies to close code, sell early, or migrate strategic control away from Europe.
3. DSF intervenes by financing companies that satisfy mission-preserving governance and technology conditions.
4. Those structural conditions improve the odds of durable, independent, European digital infrastructure.
5. Over time this produces survival, adoption, ecosystem growth, and sovereign retention rather than only isolated firm-level wins.

In one sentence:

The Digital Sovereignty Fund converts capital into sovereign digital infrastructure by financing steward-owned open-source technology companies that remain independent, durable, and European.

3.13 Step 11: Map the document's design commitments into variables

The document's qualitative commitments can now be translated directly into the quantitative model.

Document requirement	Model variable	Implication
Steward ownership	sovereignty retention d	limits extractive control over future value
Open-source licensing	openness o	keeps knowledge ordered toward shared use
EU IP retention	sovereignty retention d	retains governance in the proper community
Repayment cap discipline	r and later U	constrains rent extraction and usury pressure
Evergreen reinvestment	η and later T	recirculates capital into new productive activity

This table makes an important methodological point. The equations are not floating abstractions. They are mathematical compressions of concrete governance choices.

3.14 Step 12: Understand how impact and finance reinforce one another

The variable p appears in both the financial and impact equations:

$$M = \frac{rkp}{1 + (k-1)p}, \qquad I = N \cdot p \cdot L \cdot o \cdot d \cdot a \cdot e.$$

This shared variable is one of the most elegant features of the whole framework. It means the fund is not forced to choose between financial performance and mission if it can genuinely improve company durability. Better survival helps both.

By contrast, the repayment cap r appears directly in the financial equation but not directly in the impact equation. That asymmetry matters. It means raising r is the easiest way to improve financial output mechanically, but not necessarily the best way to improve mission output. This is exactly why theology later interrogates the composition of r.

3.15 Step 13: Translate the impact model into an operational dashboard

The framework naturally produces three classes of dashboard metrics.

Structural indicators

- percentage steward-owned companies,

- percentage open-source technology,
- percentage EU IP retention.

Operational indicators

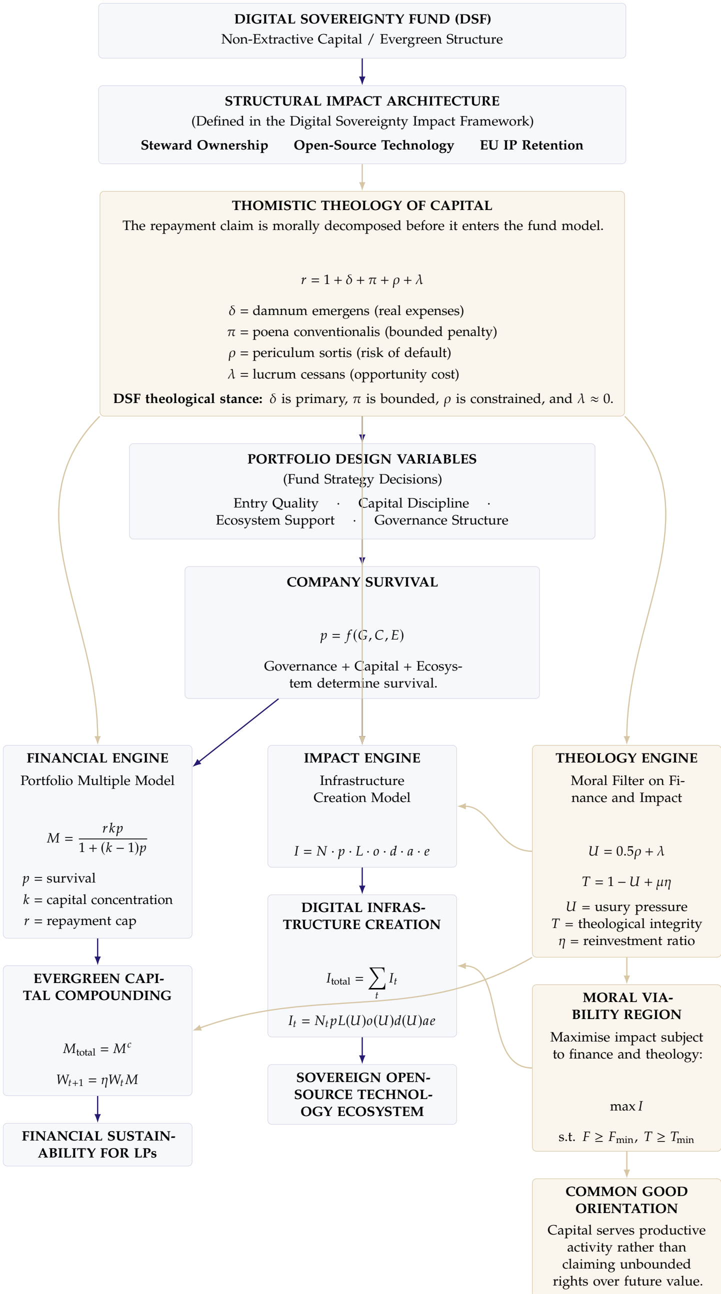
- survival rate $\to p$,
- average company lifetime $\to L$,
- organisations using portfolio technologies $\to a$,
- open-source contributors and ecosystem growth $\to e$.

Financial indicators

- portfolio multiple,
- capital recycled,
- LP distributions,
- VC benchmark comparison.

This is where the model becomes practical. The abstract variables can be tied to real reporting, real fund governance, and real accountability.

3.16 Preserved system diagram: integrated DSF architecture



3.17 Step 14: What the impact model is really claiming

At the end of the impact build-up, the reader should be able to see that the DSF theory of change is not merely "fund some good startups." Its claim is stronger:

- impact is embedded in portfolio entry requirements,
- durability matters more than headline valuation,
- sovereignty requires both openness and retained governance,
- and a surviving company matters most when it becomes a node in a wider ecosystem.

This is the conceptual bridge to the theology section. Once impact is understood structurally, the next question is whether the claims of capital strengthen or undermine that structure.

4 Thomistic Theology of Capital

4.1 The theological question

The Thomistic tradition does not reject all gain from capital. It rejects profit derived merely from the passage of time on money itself. This means the crucial question is not simply how large the repayment cap is, but what morally constitutes it.

The four extrinsic titles relevant here are:

- δ = *damnum emergens* (real expenses),
- π = *poena conventionalis* (bounded penalty),
- ρ = *periculum sortis* (risk of default),
- λ = *lucrum cessans* (opportunity cost).

Within this framework, the DSF theological stance is:

- δ is the main legitimate component,
- π may be bounded and secondary,
- ρ should be tightly constrained,
- $\lambda \approx 0$.

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The moral claim is therefore not "no return." It is "no unbounded claim on future value merely because capital could have been deployed elsewhere."

4.2 Decomposing the repayment cap

The repayment cap is decomposed as:

$$r = 1 + \delta + \pi + \rho + \lambda. \tag{19}$$

Substituting this into the financial model yields:

$$M = \frac{(1 + \delta + \pi + \rho + \lambda)kp}{1 + (k - 1)p}. \tag{20}$$

The DSF / NEC theological position can then be written as:

$$M_{\text{NEC}} = \frac{(1 + \delta + \pi + \rho)kp}{1 + (k - 1)p} \qquad \text{with } \lambda \approx 0. \tag{21}$$

This is the moment when the financial equation becomes a theological equation. The formula itself has not changed. The meaning of the repayment cap has.

4.3 Licit and usurious components

The portfolio multiple can be split into morally distinct parts:

$$M_{\text{licit}} = \frac{(1 + \delta + \pi)kp}{1 + (k - 1)p}, \tag{22}$$

$$M_{\text{usury}} = \frac{(\rho + \lambda)kp}{1 + (k - 1)p}. \tag{23}$$

This makes a crucial point visible. Two funds can show the same total multiple while differing materially in the moral composition of that multiple.

4.4 Usury pressure

The first form of the usury index is compositional:

$$U = \frac{\lambda + \rho}{r - 1}.$$

A more practical weighted form used later in the model is:

$$U = w_\rho \rho + w_\lambda \lambda \qquad \text{with } w_\lambda > w_\rho. \tag{24}$$

A concrete working specification is:

$$U = 0.5\rho + \lambda. \tag{25}$$

This is useful because not every increase in r is morally equivalent. If the repayment cap rises because real expenses rise, usury pressure need not rise. What matters is the composition of the increase.

4.5 Why capping returns matters

Return capping matters both financially and theologically. In symbolic form:

$$r \leq r_{\max}.$$

A cap means capital may be repaid and may participate modestly in success, but may not make an unbounded claim on the future value created by founders, workers, and mission.

4.6 Faith consistency and the question of "market-rate returns"

This leads to the key moral distinction:

- **same returns, same mechanisms** can amount to faith-washing,
- **same returns, different mechanisms** is the more serious claim,
- **similar long-run outcomes under a structurally different design** is also possible.

The deepest test is not whether returns are "lower" or "market-rate." It is whether the structure causes long-term value to accrue primarily to capital, or whether founders, workers, community, and mission retain most of that value.

5 Coupling Theology Back into Impact

5.1 From moral composition to mission effects

The theology does not remain an external commentary. It changes the impact equation itself.

Science Park 608 1098 XH
Amsterdam, The Netherlands
info@nonextractivecapital.com
+31 (0) 20 2621255

Let usury pressure affect durability variables:

$$L(U) = L_0(1 - \alpha U), \tag{26}$$
$$o(U) = o_0(1 - \beta U), \tag{27}$$
$$d(U) = d_0(1 - \gamma U). \tag{28}$$

Then the impact model becomes:

$$I(U) = N \cdot p \cdot L(U) \cdot o(U) \cdot d(U) \cdot a \cdot e. \tag{29}$$

This means higher extractive pressure shortens company lifetimes, weakens openness retention, and weakens sovereignty retention. Theology therefore becomes a causal variable inside the impact model.

5.2 Moral viability region

A convenient way to represent the full feasible design space is:

$$\mathcal{P} = \{(r, p, I) \mid M(r, p, k) \geq M_{\min},\ I(r, p, U) \geq I_{\min},\ U(r) \leq U_{\max}\}.$$

Traditional VC tends to move rightward by increasing r. The NEC sweet spot aims instead to move upward by increasing p while keeping r morally bounded.

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Amsterdam, The Netherlands
info@nonextractivecapital.com
+31 (0) 20 2621255

5.3 Preserved system diagram: theology interaction layer

1. MORAL DECOMPOSITION OF THE REPAYMENT CAP

$$r = 1 + \delta + \pi + \rho + \lambda$$

The key question is not only how large r is, but *what morally constitutes it*. An increase in r driven by real expenses (δ) is not the same as an increase driven by opportunity-cost claims (λ).

2A. LICIT COMPONENT

$$M_{\text{licit}} = \frac{(1 + \delta + \pi)kp}{1 + (k-1)p}$$

This is the portion most closely tied to principal, real expenses, and bounded penalties.

2B. USURY PRESSURE

$$U = 0.5\rho + \lambda$$

$$M_{\text{usury}} = \frac{(\rho + \lambda)kp}{1 + (k-1)p}$$

This measures the extractive pressure added by risk-pricing and opportunity-cost claims.

3. FINANCIAL OUTCOME

$$F = \frac{rkp}{1 + (k-1)p}$$

Finance improves if p improves, or if r rises.

4. THEOLOGICAL INTEGRITY

$$T = 1 - U + \mu\eta$$

Bounded extraction lowers U. Reinvestment (η) raises integrity.

5. IMPACT OUTCOME

$$I(U) = NpL(U)o(U)d(U)ae$$

where theology affects impact through the durability variables

6. THE COUPLING MECHANISM

$$L(U) = L_0(1 - \alpha U), \qquad o(U) = o_0(1 - \beta U), \qquad d(U) = d_0(1 - \gamma U)$$

As usury pressure rises, company lifetimes shorten, openness weakens, and sovereignty weakens. So theology is not merely commentary on the side - it changes the impact equation itself.

7. THE EVERGREEN THEOLOGICAL CLAIM

Conventional compounding looks like $W_{t+1} = W_t(1 + r_f)$, where money appears to generate more money through time alone.

The DSF evergreen structure instead says:

$$W_{t+1} = \eta W_t M$$

with each cycle tied to productive activity, capped repayment, and reinvestment into new productive activity.

In other words: capital compounds through the repeated creation of real goods, not through time alone.

5.4 Scenario table: same repayment cap, different moral composition

The following scenario table preserves a key insight from the earlier modelling work: the same nominal repayment cap can generate the same financial multiple while differing strongly in theological composition and mission consequences.

Assume:

- $N = 40$,
- $p = 0.60$,
- $k = 5$,
- $L_0 = 8$,
- $o_0 = 1.0$,
- $d_0 = 1.0$,
- $a = 2.0$,
- $e = 1.2$,
- $L(U) = 8(1 - 0.20U)$,
- $o(U) = 1 - 0.15U$,
- $d(U) = 1 - 0.20U$.

Scenario	δ	π	ρ	λ	r	$U = 0.5\rho + \lambda$	M	**Impact tendency**
A. Strongly licit	1.60	0.20	0.10	0.00	2.90	0.05	2.49	Very high
B. Mostly licit	1.30	0.20	0.30	0.10	2.90	0.25	2.49	High
C. Mixed	0.90	0.20	0.50	0.30	2.90	0.55	2.49	Moderate
D. Extractive drift	0.60	0.10	0.70	0.50	2.90	0.85	2.49	Lower
E. Highly usurious	0.30	0.10	0.80	0.70	2.90	1.10	2.49	Much lower

Approximate impact comparison under those assumptions:

Scenario	U	$L(U)$	$o(U)$	$d(U)$	**Relative impact** I
A. Strongly licit	0.05	7.92	0.9925	0.99	~ 451
B. Mostly licit	0.25	7.60	0.9625	0.95	~ 417
C. Mixed	0.55	7.12	0.9175	0.89	~ 351
D. Extractive drift	0.85	6.64	0.8725	0.83	~ 289
E. Highly usurious	1.10	6.24	0.8350	0.78	~ 244

A second preserved comparison shows how rising repayment caps can drift from licit to extractive structures:

Scenario	r	Example composition	U	Financial effect	Impact effect
Low cap, licit	2.0	mostly δ	low	modest	strong
Moderate cap, licit	2.5	mostly δ, small ρ	low-moderate	better	still strong
Higher cap, mixed	3.0	more ρ, some λ	moderate	better	weaker
High cap, extractive	4.0	substantial λ and ρ	high	high	much weaker
VC-like logic	8.0+	dominated by λ and ρ	very high	extreme upside	mission distortion

The key takeaway is that moral legitimacy is not determined only by the size of the repayment cap. It is determined by the composition of that cap.

6 Tri-Objective Optimisation

To model the tension explicitly, define:

$$F = \text{financial performance}, \qquad I = \text{impact performance}, \qquad T = \text{theological integrity}.$$

The financial layer is:

$$F = M = \frac{rkp}{1 + (k-1)p}.$$

The impact layer is:

$$I = N \cdot p \cdot L(U) \cdot o(U) \cdot d(U) \cdot a \cdot e.$$

The theological layer is:

$$T = 1 - U + \mu\eta. \tag{30}$$

This makes the trade-off legible:

$$\frac{\partial F}{\partial r} > 0$$

while often

$$\frac{\partial T}{\partial r} < 0$$

if the increase in r is being driven by ρ and λ.

And because higher usury pressure harms durability variables,

$$\frac{\partial I}{\partial U} < 0.$$

So the DSF objective is best written as:

$$\max I \quad \text{subject to } F \geq F_{\min} \text{ and } T \geq T_{\min}. \tag{31}$$

This is more precise than saying "maximize returns" or even "maximize blended value." It says the fund should maximize infrastructure creation, but only inside a region where financial viability and moral legitimacy are both preserved.

7 Evergreen Compounding and the Theology of Capital

7.1 Three compounding logics

The evergreen model becomes clearest when contrasted with two other compounding structures.

1. Interest-based compounding

$$W_{t+1} = W_t(1 + r_f), \qquad W_t = W_0(1 + r_f)^t.$$

Key property: growth depends on capital, time, and interest rate. Production is not structurally required.

2. Venture-capital compounding

$$W_{t+1} = W_t M, \qquad M = \frac{rkp}{1 + (k-1)p}.$$

Key property: the model tends to maximize r through rare outsized outcomes and aggressive extraction.

3. Evergreen stewardship compounding

$$W_{t+1} = \eta W_t M.$$

Here returns are recycled into new productive activity rather than distributed through a terminal exit logic alone.

7.2 Why evergreen compounding matters theologically

The theological problem with classical interest compounding is that money appears self-generating. The evergreen structure changes this logic.

The DSF claim is:

- returns come from productive companies,
- returns are bounded,
- capital is recycled into new productive activity.

Impact across cycles is then:

$$I_{\text{total}} = \sum_{t=1}^{c} I_t.$$

The theological integrity function includes reinvestment explicitly:

$$T = 1 - U + \mu\eta.$$

So theology interfaces with evergreen compounding in two ways:

- low U means bounded extraction,
- high η means high circulation of capital into new productive goods.

This leads to the deepest summary sentence of the model:

In conventional finance, capital compounds through time or financial exits. In an evergreen stewardship model, capital compounds through the repeated creation of productive enterprises and sovereign digital infrastructure.

8 Unified Interpretation

The unified logic of the DSF model can be stated in one progressive chain:

$$\text{capital} \longrightarrow \text{companies} \longrightarrow \text{infrastructure} \longrightarrow \text{ecosystem} \longrightarrow \text{sovereignty} \longrightarrow \text{reinvestment}.$$

Compared with a conventional VC logic,

$$\text{capital} \longrightarrow \text{companies} \longrightarrow \text{exits} \longrightarrow \text{returns},$$

the DSF model adds three decisive transformations:

1. it turns impact into a structural property of portfolio entry,
2. it morally decomposes the repayment claim of capital,
3. it treats compounding as recirculated productive stewardship rather than as money breeding money.

The core financial strategy is therefore not to inflate r without limit. It is to improve p, preserve L, o, and d, cap extraction, and recycle capital into new common goods.

Science Park 608 1098 XH
Amsterdam, The Netherlands
info@nonextractivecapital.com
+31 (0) 20 2621255

A Legend of Variables

A.1 Financial and portfolio variables

Variable	Meaning	Role in the model
M	Single-cycle portfolio multiple	Core financial outcome for one investment cycle.
M_{total}	Multi-cycle / evergreen multiple	Extends the single-cycle result across c cycles.
F	Financial outcome	Used in the optimization constraint $F \geq F_{\min}$.
p	Survival or success probability	Shared variable linking finance and impact.
k	Capital concentration	Measures how much more capital is allocated to successful firms.
r	Repayment cap multiple	Total repayment claim made by the fund.
c	Number of evergreen cycles	Counts how many times capital is recycled.
W_t	Fund wealth at time t	Tracks evergreen capital over time.
W_{t+1}	Next-period wealth	Computed after repayment and reinvestment.
r_f	Conventional interest rate	Used only as a contrast case in standard interest compounding.
$F_{\min}$	Minimum acceptable financial threshold	Lower bound for financial viability.
N	Number of projects or companies funded	Basic scale of the portfolio.
S	Successful companies	Used in the pre-probability formulation of the portfolio model.
F and S with N	Failures and successes	Used in the asymmetric deployment derivation before conversion to p.
I_f	Investment in a failure-stage company	Base deployment level.
I_s	Investment in a successful company	Equal to kI_f.
T	Average repayment time in years	Used in the cash-flow extension.
C_{ops}	Annual operating cost	Salary plus administration.
C_{mgmt}	Management fee burden	Fee extension in the cash-flow model.
L in the loan formula	Loan principal	Simplified notation for the NVP loan amount.
i	Loan financing rate	Used in the cash-flow extension.
T_L	Loan horizon	Amortisation period in the cash-flow extension.
d_{LP}	LP distribution fraction	Share of annual cash distributed to investors.

A.2 Impact and infrastructure variables

Variable	Meaning	Role in the model
I	Impact outcome	Main infrastructure-creation measure.
I_t	Impact in period t	One time-slice of infrastructure creation.
I_{total}	Total impact across periods	Sum of impact across the evergreen horizon.
L	Company lifetime	Duration that a successful firm remains active.
L_0	Baseline company lifetime	Starting lifetime before theological pressure is applied.
o	Openness retention factor	Measures whether the firm remains open-source.
o_0	Baseline openness retention	Starting openness before theological pressure is applied.
d	Sovereignty retention factor	Measures whether governance and control remain in Europe.
d_0	Baseline sovereignty retention	Starting sovereignty before theological pressure is applied.
a	Adoption factor	Captures actual uptake, use, or deployment of the technology.
e	Ecosystem spillover multiplier	Captures reuse, collaboration, spinouts, and ecosystem effects.
G	Governance design	One input into the survival function $p = f(G, C, E)$.
C	Capital structure or discipline	One input into the survival function $p = f(G, C, E)$.
E	Ecosystem support	One input into the survival function $p = f(G, C, E)$.

A.3 Theology, usury, and moral-constraint variables

Variable	Meaning	Role in the model
δ	*Damnum emergens*	Real expenses or real costs legitimately borne by the fund.
π	*Poena conventionalis*	Bounded penalty or late-payment component.
ρ	*Periculum sortis*	Risk-of-default component.
λ	*Lucrum cessans*	Opportunity-cost claim; morally the most suspect component here.
U	Usury-pressure index	Measures extractive moral pressure from ρ and λ.
T	Theological integrity	Summary measure of moral acceptability.
$T_{\min}$	Minimum theological threshold	Lower bound for theological viability in the optimization region.
η	Reinvestment ratio	Fraction of returns recycled into new productive activity.
μ	Reinvestment-weight parameter	Measures how strongly reinvestment raises theological integrity.
M_{licit}	Licit component of the portfolio multiple	Portion attributed mainly to principal, real expenses, and bounded penalties.
M_{usury}	Usury-linked component of the portfolio multiple	Portion attributed to risk-pricing and opportunity-cost claims.
α	Lifetime sensitivity to usury pressure	Governs how quickly $L(U)$ falls as U rises.
β	Openness sensitivity to usury pressure	Governs how quickly $o(U)$ falls as U rises.
γ	Sovereignty sensitivity to usury pressure	Governs how quickly $d(U)$ falls as U rises.
$\mathcal{P}$	Moral viability region	Set of portfolio designs that satisfy finance, impact, and moral constraints simultaneously.

B One-Sentence Unified Logic

The Digital Sovereignty Fund converts capital into sovereign digital infrastructure by financing steward-owned open-source companies that remain European and durable, while morally constraining the claims of capital through capped, evergreen, and theology-shaped finance.